Estimating Expected Returns (Mean) II

Recap: Expected Return (Mean)

$$E[r_j] = \frac{1}{n}\sum_{t=1}^{n} r_j$$

Where:

$E[r_j]$ = Expected return on a stock *j*

r_j = Realised return on stock *j*

n = Number of return observations for *j*.

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Recap: Calculating Returns

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$$E[r_j] = \frac{1}{n}\sum_{t=1}^{n} r_j$$

In Python:

```
# Store the expected return (mean) as a
# variable called "expected_return"
expected_return = df['returns'].mean()
```

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Recap: Annualising $E[r]$

The crude method:

$$E[r_{ANNUAL}] = E[r_{DAILY}] \times 250$$

The sophisticated method:

$$E[r_{ANNUAL}] = (1 + E[r_{DAILY}])^{250} - 1$$

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We can create custom ‘functions’ on Python to do a lot of work for us.

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Python Walkthrough

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Summary

The easiest way to calculate the ‘expected return’ on a stock is to compute its historic average return.

Daily average expected returns can be annualised by:

- The ‘crude’ method: $E[r_{Daily}] \times 250$
- The ‘sophisticated’ method: $\left(1 + E[r_{Daily}]\right)^{250} - 1$

We can implement these calculations in a ‘function’ on Python using:

```
def functionName(param1, param2, ... , paramN):
    # Code for the actual function.
```

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Now have a go at the quiz!

